

Formula 1 Ratings and Market Efficiency

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Abstract

We compare rating systems on the full history of Formula 1 (1,158 races, 1950 to present). Grids of up to fifty-five cars break the pairwise decompositions used by standard systems: Glicko-2 scores below a uniform baseline. A Thurstonian lattice system, which computes joint order statistics exactly and accepts any performance noise distribution, outperforms TrueSkill, Elo, Glicko-2 and OpenSkill, and improves further when its noise density includes a separated block of slow mass matching the retirement process. We then attempt the harder task of testing market efficiency with the incomplete odds record that survives in web archives (135 races). The market leads the best ratings by 0.57 of log loss and a fitted combination gives the ratings zero weight.

1 Introduction

A rating system assigns each competitor a skill estimate, updates it after every contest, and predicts future outcomes. Most systems in use were designed for two-player games and extend to multi-entrant events by decomposition: Elo, Glicko-2 and the Weng–Lin systems treat a finish order as a collection of pairwise results, and TrueSkill chains adjacent finishers. For small fields the decomposition costs little. Formula 1 provides fields of ten to fifty-five cars, where a single race implies up to 190 correlated pairwise comparisons, and we will see that the standard decompositions do not merely lose accuracy there but can fail outright. The primary purpose of this paper is to compare rating systems on this terrain and to explain why a Thurstonian treatment, meaning one that works with the joint distribution of the full finish order, is essential to doing well on it.

Formula 1 also poses an unusual distributional problem. The fraction of entrants failing to finish, by decade:

decade	1950s	60s	70s	80s	90s	2000s	10s	20s
DNF rate	.42	.49	.51	.59	.51	.30	.18	.13

For most of the sport’s history, retirement was the most likely outcome of starting a race. A rating model committed to Gaussian performance noise has no way to represent this; a model whose noise density is a free input does. Section 4 exploits this freedom.

A secondary question is whether the resulting ratings can beat the betting market. We emphasize that with the data available this is a hard test to run at all: no public archive of Formula 1 winner odds exists, and the record we recover from web archives is incomplete and unevenly timed. Section 6 assembles that record and Section 7 reports the comparison, which favors the market decisively.

2 Data and protocol

Race results come from the community-maintained fldb compilation (CC-BY), covering 1,158 championship races from 1950 through the 2026 season. Drivers completing at least 90% of the race distance receive official positions even if they stop early. Other retirements carry no position and enter our protocol tied at the back of the field. Shared drives in the 1950s produce genuine dead heats, which are treated as ties.

Evaluation is prequential. Races are processed in date order; the first 20% are unscored warm-up; each later race is predicted before it is observed. We report log loss, Brier score and accuracy for the winner prediction, Kendall’s tau between predicted and observed orders, expected calibration error (ECE) over pooled win probabilities, and a rank-PIT statistic that measures whether the predicted distribution over finish positions matches the realized positions.

3 Rating systems compared

The Thurstonian system used here is built on a discrete lattice of ability values. Each driver’s belief is a probability density on the lattice, unrestricted by any parametric family. A race performance is ability plus noise, with the noise drawn from a base density that is an input to the system. Between races, beliefs diffuse in proportion to elapsed time.

Prediction is a forward computation: each driver’s predictive performance density is the belief convolved with the base density, and win probabilities are order statistics of the field, computed exactly on the lattice, ties included. Updating is a backward computation: given an observed finish order, the likelihood of the whole order as a function of one driver’s ability is computed exactly, holding opponents at their predictive densities. Index finishers from fastest to slowest. A forward pass computes, at each performance level, the probability that finishers 1 through $k - 1$ performed better in the observed order; a backward pass does the same for finishers $k + 1$ through N performing worse. The likelihood for the k -th finisher is the base density correlated with the product of the two passes. The update costs $O(N)$ lattice operations per race and was validated against brute-force simulation.

The benchmark below runs all systems over identical data under the protocol of Section 2. TrueSkill is configured with the empirical dead-heat rate, and each package uses its own prediction formula.

system	log loss	accuracy	tau	ECE	rank-PIT
Thurstone lattice	2.225	.316	.328	.015	.176
TrueSkill	2.346	.275	.300	.011	.199
Elo (multi-entrant)	2.465	.303	.319	.026	.220
OpenSkill Plackett–Luce	2.705	.259	.283	.023	.223
OpenSkill Thurstone–Mosteller	2.951	.007	.106	.010	.628
uniform baseline	3.162	.043	—	—	—
Glicko-2	3.727	.271	.255	.022	.166

The joint treatment leads on log loss, accuracy and rank correlation. The failures are the instructive part. Glicko-2 scores below the uniform baseline: converting a twenty-car grid into nineteen results within one rating period destabilizes its volatility estimation, and our implementation already includes stabilizing guards without which it does not complete the dataset.

OpenSkill’s all-pairs Thurstone–Mosteller variant diverges the same way. These systems are sound at head-to-head scale; what breaks them here is treating 190 correlated comparisons as independent evidence. This is why exact joint order statistics are not a refinement in this setting but a requirement.

4 Choosing the noise distribution

Every comparison system above is committed to Gaussian beliefs with Gaussian or Gumbel noise. The lattice system accepts any base density, so the noise model becomes an empirical question, and the retirement table in Section 1 suggests the answer should not be Gaussian.

The motivated candidate is a two-component density: with probability $1 - p$ the noise is standard Gaussian, and with probability p it is drawn uniformly from a wide interval on the slow side. Figure 1 shows the shape. The rectangular block represents retirement: a mechanical failure produces a result unrelated to pace, so its probability mass is separated from the pace distribution rather than blended into its tail.

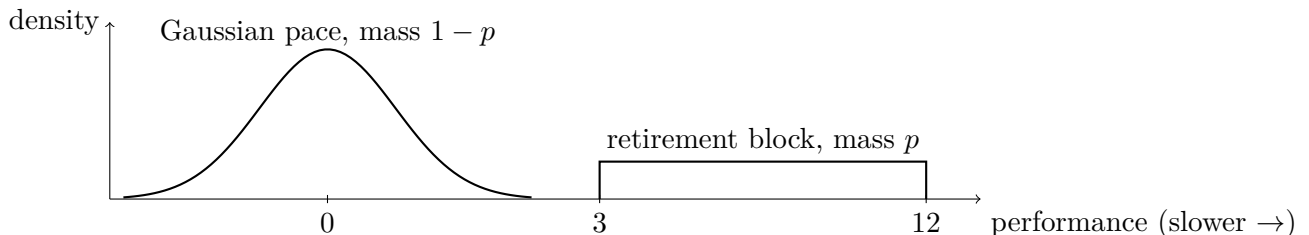


Figure 1: The disaster-slab base density: a continuous Gaussian pace component and a separated rectangular block of slow mass representing retirement.

base density	log loss	ECE	rank-PIT
era-adaptive slab	2.200	.016	.185
$N(0, 1)$ plus 25% slab	2.203	.015	.158
skew-normal ($a=2$)	2.222	.010	.165
Gaussian	2.225	.015	.176
tight core (0.5) plus 25% slab	2.254	.013	.136
Student- t ($\nu=5$)	2.271	.018	.171
near-Dirac core plus slab	2.42–2.52	—	—

The slab improves every metric relative to the Gaussian. Because a fixed 25% mass is wrong for most decades, the era-adaptive variant estimates p from a trailing window of observed retirement frequency, using past races only. It attains the best log loss and rank correlation of any configuration tested, and its fitted mass declines across the dataset to finish near the modern retirement rate.

The failed alternatives are informative. Symmetric heavy tails (the t family) hurt increasingly as tails fatten, because symmetry places equal mass on implausibly fast performances. Skew-normal noise changes little, consistent with two facts about rank data: the difference of two independent identically distributed noises is symmetric whatever their skew, and ranks are invariant to monotone transformations of the time scale. A near-Dirac pace core loses about

0.2 of log loss, showing that variation from strategy, traffic and weather is comparable in size to ability differences. None of these choices is available to a system committed to a parametric family, which is the second reason the Thurstonian treatment matters here.

5 Summary of the ratings comparison

Two properties carried the result: exact joint order statistics, without which systems degrade or diverge at grid scale, and a free noise density, which converts knowledge of the retirement process into forecast accuracy. Both are structural. They cannot be patched onto a pairwise Gaussian system by tuning.

6 The odds record

We know of no public archive of Formula 1 winner odds, so we assembled what survives. The Wayback Machine holds captures of odds-comparison pages; 523 usable captures yield full-field win odds for 144 races between 2008 and 2025, each covering roughly twenty bookmakers. For each race we take the median implied probability per driver across bookmakers and normalize within the race, which removes the bookmaker margin. The record is incomplete and unevenly timed: 2011–2012 and 2017–2020 are sparse, many captures postdate qualifying, and a few precede the race by more than a week. These limitations should be kept in view; they make the efficiency test below possible but imperfect. We publish only the derived consensus probabilities.

7 The efficiency test

On the 135 covered races falling after the warm-up period, with market and ratings scored on the same matched set of drivers:

	log loss
market consensus	1.547
fitted pool ($a=1.10$, $b=0.00$)	1.538
ratings given all past odds	1.898
era-adaptive slab ratings	2.117
uniform	3.018

The market leads the best ratings by 0.570 of log loss (standard error 0.065). A pool of the form $q \propto m^a s^b$, with exponents fitted on past races only, assigns the ratings weight $b = 0$. The result survives enriching the challenger: feeding the ratings the consensus odds of all previous races as additional observations improves their standalone score from 2.117 to 1.898, but the fitted weight remains zero. Prices subsume everything the ratings know.

The explanation is straightforward. Post-qualifying prices incorporate grid position, current car performance and weather, none of which is visible to a system built from past results. The one detectable miscalibration is the fitted exponent $a = 1.10$, indicating slight underconfidence about favorites, in the direction of the favorite-longshot bias documented in other racing markets. It is worth 0.008 of log loss, well inside any bookmaker margin. Within the limits of the archived record, Formula 1 winner markets appear efficient with respect to results-based ratings, including ratings augmented with the market’s own price history.

Two qualifications. First, snapshot timing favors the market, since post-qualifying information is present in many captures; closing-price data would sharpen the estimate but is unlikely to reverse a gap of nine standard errors. Second, the test covers winner markets only. Book-makers price derivative markets such as head-to-heads and top-six finishes from winner odds using simplifying rank models, and our results on rank models suggest that is where pricing error would concentrate; testing it requires derivative odds the archives do not contain.

8 Limitations

Constructor effects are folded into driver ratings; separating driver from car is the natural next study. Retirements enter tied at the back, which discards the official ordering of retirees by laps completed. The ratings are one-dimensional, with no axis for wet weather or circuit type. The odds record is as described in Section 6.

Reproducibility

Each table is one command from the `winning` package (github.com/microprediction/winning): `python -m winning.benchmarks.run_benchmark --dataset f1` for Section 3, and the scripts `research/f1_dirac_disaster.py`, `research/t_sweep.py`, `research/f1_era_slab.py`, `research/f1_odds_fe`, `research/f1_odds_parse.py` and `research/f1_market_test.py` for Sections 4 through 7. Race data is fetched at run time from `f1db` (CC-BY-4.0). The lattice machinery is the `thurstone` package; the underlying ability transform is described in Cotton, *SIAM J. Financial Mathematics* 12(1), 2021.